

More Dietetic Time, Better Outcome?

A Randomized Prospective Study Investigating the Effect of More Dietetic Time on Phosphate Control in End-Stage Kidney Failure Haemodialysis Patients

Belinda Morey^a Rebecca Walker^a Andrew Davenport^b

^aDepartment of Nutrition and Dietetics, Royal Free Hospital, and ^bUCL Centre for Nephrology, Royal Free and University College Hospital Medical School, London, UK

Key Words

Compliance, phosphate control · Dietitian · Hyperphosphataemia · Haemodialysis, dietetic time · Nutrition · Phosphate

Abstract

Background/Aim: A randomized controlled trial was conducted involving 67 stable adult haemodialysis patients with hyperphosphataemia. The objective was to determine the effect of monthly dietetic consultations on patients' serum phosphate concentrations and calcium × phosphate product. **Methods:** The intervention group received monthly dietetic consultations for 6 months using advanced counselling skills aimed at limiting phosphate intake in the diet and improving compliance with phosphate binders, whereas standard care with dietetic consultations every 6 months was provided to the control group. **Results:** Serum phosphate concentrations decreased in the intervention group from 2.05 ± 0.48 to 1.80 ± 0.48 mmol/l by month 3 ($p < 0.05$). However, this subsequently increased and by month 6 there was no significant difference from baseline. After controlling for confounding variables, the difference between the groups was approaching significance at month 3, but there was no difference by month 6, or following conclusion of the study. The calcium × phosphate product similarly decreased in both groups, achieving K/DOQI standards, but was not sustained either during the remainder of the study but decreased further by the 12-month follow-up and, although it was not significantly different, it did improve sig-

nificantly from baseline in both the intervention and control groups. **Conclusion:** Increased frequency of dietetic consultations can result in improved phosphate control in the short term, and therefore more innovative strategies appear necessary to sustain longer-term control.

Copyright © 2008 S. Karger AG, Basel

Introduction

It is well recognized that high serum phosphate concentrations in end-stage kidney failure patients treated by both haemodialysis and peritoneal dialysis are associated with an increased risk of both morbidity and mortality [1–3]. A raised serum phosphate concentration is associated with the development and worsening of hyperparathyroidism, renal osteodystrophy, and vascular and cardiac calcification [2–7]. Phosphate retention and elevated serum phosphate concentrations result from progressive chronic kidney disease, due to the reduction in glomerular filtration rate and derangement of hormones involved in the complex mechanisms of phosphate metabolism. In established dialysis patients, the amount of phosphate cleared during each haemodialysis session and similarly with peritoneal dialysis exchanges is typically less than that absorbed from the diet, and hyperphosphataemia is then often exacerbated by non-compliance with dietary restrictions and phosphate binder medications [8, 9].

Clinical Trials registration NCT00454727.

KARGER

Fax +41 61 306 12 34
E-Mail karger@karger.ch
www.karger.com

© 2008 S. Karger AG, Basel
1660–2110/08/1093–0173\$24.50/0

Accessible online at:
www.karger.com/nec

Rebecca Walker
Department of Nutrition and Dietetics, Royal Free Hospital
Pond Street
London NW3 2QG (UK)
Tel. +44 207 599 1719, Fax +44 207 599 1718, E-Mail Rebecca.Walker@royalfree.nhs.uk

A systematic review of treatment compliance with medications by the Cochrane group [10] suggested that people who are prescribed self-administered medications (short- and long-term treatments) typically take less than half the prescribed dose. End-stage kidney failure patients on dialysis are often prescribed a multitude of medications, including vitamins, antihypertensives, and other cardioprotective medications, as well as phosphate binders. It is therefore not surprising that compliance with phosphate binders varies between patients. To try and overcome non-compliance, several studies have reported that combinations of more complex interventions, such as convenient care, reminders, reinforcement, counselling, and supportive care, amongst others, can result in small improvements in long-term adherence and health outcomes, and this lifestyle change approach has been successfully used in the management of diabetics and cardiovascular disease patients.

Durose et al. [11] investigated whether patient general knowledge about phosphate and also more specifically knowledge about which foodstuffs contained high phosphate levels affected compliance. The authors showed that dietary knowledge of foodstuff phosphate content was not predictive of dietary compliance and that to improve compliance, additional educational techniques, which focus on motivating patients to make changes, was required. Ashurst Ide and Dobbie [12] investigated the effect of additional education on compliance with phosphate binders, and showed that in-depth counselling, using an education tool, with a focus on both knowledge and compliance with diet and medications produced a reduction in serum phosphate concentrations of 0.36 mmol/l, which was sustained for 3 months following the educational intervention programme.

This is supported by other investigators [13] who have also reported that regular contact and dietary re-enforcement can also be beneficial. However, not all studies have reported that additional time spent with patients resulted in improved compliance and/or reduction in phosphate concentrations. One such study [14] noted that behavioural techniques subsequently can fail when contact with the interventionist becomes less frequent or absent.

The renal dietician plays a pivotal role in serum phosphate management, advising patients on which foods to restrict and how to match the dose of phosphate binders to the phosphate content of a meal, whilst at the same time considering the fine balance of restricting phosphate intake whilst maintaining sufficient dietary caloric and protein intake to prevent malnutrition [8, 15].

As the results of previous studies show a variable effect of additional dietetic intervention on patients' serum phosphate control, we performed a randomized prospective study, designed to examine the influence of monthly dietetic consultations on patients' serum phosphate concentrations, compared to our local standard policy of a dietetic review every 6 months. The aim of this study was to determine if additional intervention in a group of haemodialysis patients with persistent hyperphosphataemia would be successful in achieving target phosphate control and also whether any effect would be sustained.

Patients and Methods

Serum phosphate concentrations were reviewed over 3 months (from September to November 2005) for all haemodialysis patients ($n = 482$) associated with a London tertiary referral hospital and satellite units. Inclusion criteria were a mean serum phosphate level persistently above the UK Renal Association Standards Committee target <1.8 mmol/l [16], being on maintenance haemodialysis >6 months, age >18 years, ability to give informed consent, a stable urea reduction rate $>65\%$, and a body mass index >20 . Patients were excluded if they had lost $>7.5\%$ dry weight in the previous 3 months, if they had alcohol/drug dependency, a solid-organ malignancy, total parathyroidectomy, a parathyroid hormone level >30 pmol/l in conjunction with a corrected serum calcium concentration >2.7 mmol/l, or prescribed cinacalcet. In addition those who were unable to provide appropriate informed consent due to mental illness/learning difficulties/dementia or if dialysis adequacy dropped were also excluded.

The Whittington and Moorfields Research Ethics Committee and the Royal Free Hospital Trust Local Research Ethics Committee approved the trial.

A total of 75 patients fulfilled the inclusion criteria and 67 provided informed consent to take part. These patients had persistently failed to achieve the UK Renal Association Standards Committee target serum phosphate level [16] and many were known to be non-compliant with previous standard dietetic care. The primary investigator was blinded to the randomization procedure (random number generation) and the characteristics of subjects in each of the groups are outlined in table 1. During the study, 7 patients were excluded for various reasons (fig. 1).

All of the consented subjects were reviewed individually, whilst on dialysis, at baseline/month 1 by a specialist renal research dietician prior to randomization. Those allocated to the intervention group were then reassessed regularly, each month for a 6-month period, whereas those in the control group were not reviewed again until month 6 (locally agreed level of service for persistent hyperphosphataemia). The dietician assessed subjects' diets using a diet history and approximation of dietary phosphate content and nutritional adequacy, as well as phosphate binder compliance by self-report against prescription.

Subjects were advised and educated about dietary phosphate restriction and compliance with phosphate binder prescription while maintaining nutritional adequacy and a variety of strategies were employed to encourage dietary modification including

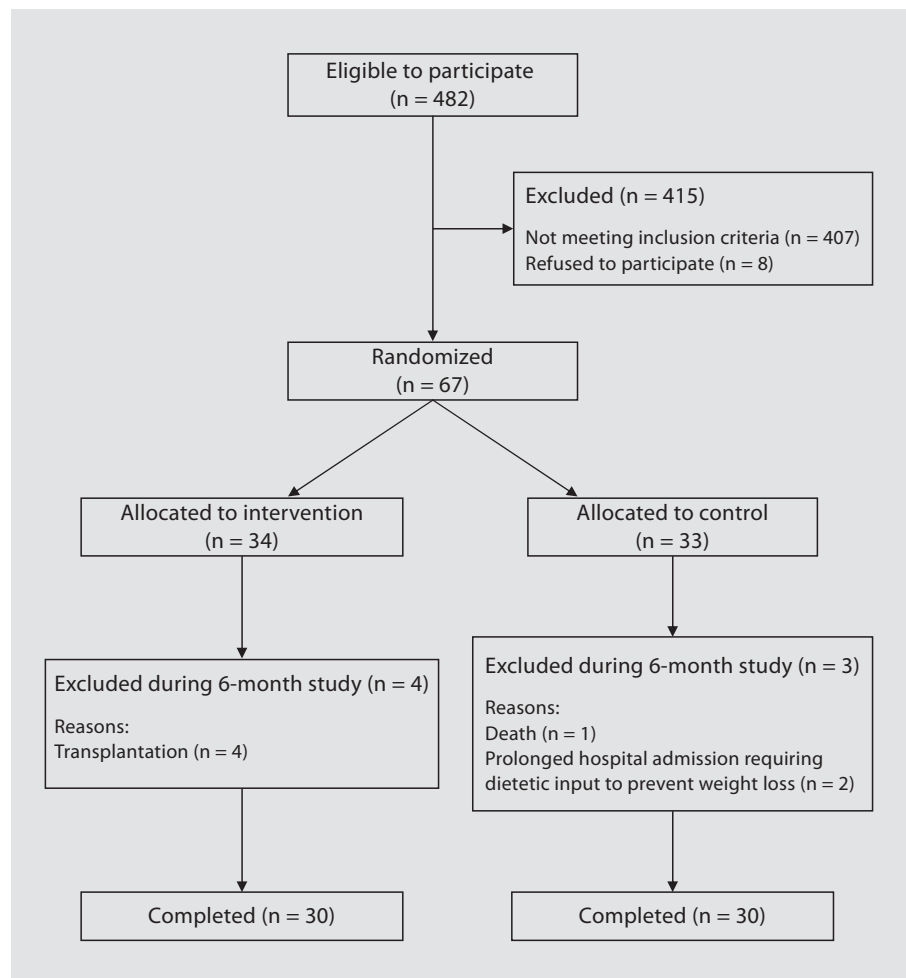


Fig. 1. Flowchart of participant progression.

Table 1. Characteristics of subjects at baseline (no significant differences between the groups)

	Intervention group (n = 34)	Control group (n = 33)
Age, years	60.4 ± 15.6	54.9 ± 15.9
Sex, % male	73.5	51.5
Ethnicity		
White, %	52.9	48.5
Indo-Asian, %	20.6	15.2
Black, %	14.7	30.3
Other, %	11.8	6.1
Requiring interpreter, %	5.9	21.2
Months since commencement of dialysis (range)	45 (7–327)	41 (6–283)
Dialysis prescription (three times a week), hours (range)	4 (3.5–5)	4 (3.5–6)
Body mass index	26.9 (19.4–37.5)	24.9 (19.2–43.4)
Grip strength, kg	22.3 ± 9.2	20.7 ± 10.7
Mid-upper-arm circumference, cm	29.0 ± 3.2	28.9 ± 4.1

Table 2. Biochemical characteristics of subjects at baseline (no significant differences between the groups)

	Intervention group (n = 34)	Control group (n = 33)
Phosphate, mmol/l	2.05 ± 0.5	2.24 ± 0.5
Potassium, mmol/l	5.3 ± 0.70	5.2 ± 0.8
Corrected calcium, mmol/l	2.28 ± 0.20	2.31 ± 0.16
Calcium × phosphate product, mmol ² /l ²	4.66 ± 1.08	5.20 ± 1.42
Pre-dialysis urea, mmol/l	21.6 ± 5.6	23.1 ± 5.1
Albumin, mmol/l	41.1 ± 3.4	40.4 ± 3.1
Parathyroid hormone, pmol/l (range)	38.7 (1.9–81.4)	33.6 (4.1–142.)
Phosphate binder type, %		
Sevelamer hydrochloride	11.8	12.1
CaCO ₃	26.5	30.3
Calcium acetate	2.9	0.00
Aluminium based	5.9	9.1
CaCO ₃ and aluminium based	5.9	3.0
Sevelamer hydrochloride and aluminium based	8.8	15.2
Sevelamer hydrochloride and CaCO ₃	29.4	21.2
Sevelamer hydrochloride and calcium acetate	2.94	3.03
Sevelamer hydrochloride, CaCO ₃ , and aluminium based	5.88	3.03
Sevelamer hydrochloride, CaCO ₃ , aluminium based, and MgCO ₃	0.00	3.03
Prescribed dose of binders, number of tablets/day (range)	9 (2–20)	9 (2–21)
Self-reported dose of binders taken, number of tablets/day (range)	7 (2–18)	7 (0–21)
Prescribed dose of alphacalcidol, µg/week (range)	2 (0–15)	3 (0–12)

motivational counselling, negotiation, behaviour modification therapy, reminders, reinforcement, supportive care, and both written and verbal education. The research dietician individualized strategies to each subject. Patients were educated on how to match phosphate binders to the phosphate content of a meal. The dietician also liaised with the medical team to adjust phosphate binder prescriptions to better match the needs of the individual patient. Following the 6-month trial period, standard care was recommenced for all subjects seeing their regular unit dietician if referred.

SPSS was used for all analyses and $p < 0.05$ was regarded as significant. All continuous variables were tested for normal distribution; only parathyroid hormone showed a skewed distribution. PO₄ levels were tested using paired t tests, independent t tests, analysis of covariance, and Wilcoxon signed-rank test. χ^2 tests were used to assess compliance with phosphate binders. The data from the excluded subjects (reasons displayed in figure 1) were included in the analysis until the time point of exclusion.

Results

Characteristics of Participants

Characteristics of subjects in both the intervention and control groups were similar at baseline, as shown in table 2.

Primary Outcomes

The mean serum phosphate concentration for the intervention group initially declined, from a baseline of 2.05 ± 0.48 mmol/l, with the maximum effect observed at month 3, 1.80 ± 0.48 mmol/l ($p = 0.003$), but thereafter stabilized and was not statistically different from the baseline concentration by month 6, at 1.90 ± 0.43 mol/l ($p = 0.081$; fig. 2). As for the control group, there was no statistically significant change in serum phosphate concentrations during the study period. After the study had been completed, there was no difference in serum phosphate concentrations between the groups, both at 3 months (fig. 2) and also some 12 months later: in 27 patients of the intervention group 1.64 ± 0.42 mmol/l vs. 1.86 ± 0.54 mmol/l in 21 patients of the control group.

Using analysis of covariance and adjusting for potential confounding variables, including baseline serum phosphate concentrations, use of aluminium-based binders at baseline (well recognized as the most potent type of binder), and dose of alphacalcidol, the adjusted difference in means at month 3 was -0.253 ($p = 0.056$, 95% confidence interval -0.513 to 0.007). At month 6, there was no significant difference between the mean serum phosphate concentrations for the intervention group com-

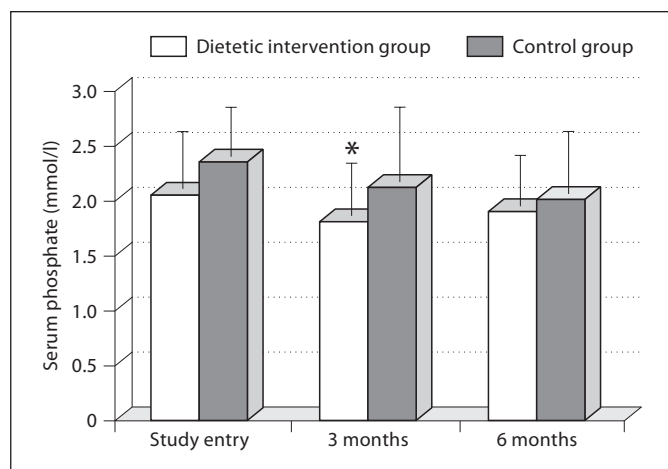


Fig. 2. The serum phosphate concentration initially declined in the intervention group, with the greatest reduction at month 3 (1.80 mmol/l compared to baseline 2.05 mmol/l; * $p = 0.003$), but by month 6 (1.90 mmol/l) was not significantly different from baseline ($p > 0.05$).

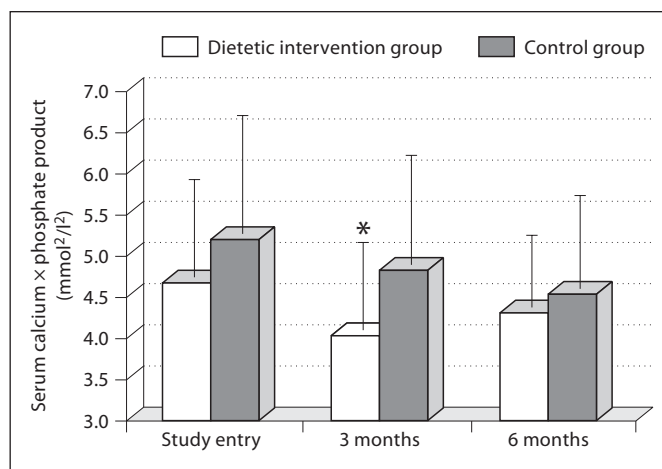


Fig. 3. The serum calcium $\times$ phosphate product declined in both groups, with the greatest difference (* $p < 0.05$) between the groups seen at month 3 (intervention baseline, 4.66 mmol²/l², month 3, 4.04 mmol²/l², and month 6, 4.30 mmol²/l² versus control baseline, 5.20 mmol²/l², month 3, 4.82 mmol²/l², and month 6, 4.55 mmol²/l²).

pared to the control group; similarly, there were no statistical differences on follow-up after the trial had ended, for up to 12 months.

In terms of the calcium $\times$ phosphate product (fig. 3), reductions were observed in both groups during the trial period. In the intervention group, although the mean product declined from 4.66 ± 1.08 mmol²/l² at baseline to 4.30 ± 0.94 mmol²/l² by month 6, this was statistically not significant. However, in the control group, the mean calcium $\times$ phosphate product fell from 5.20 ± 1.42 to 4.55 ± 1.05 mmol²/l², which was just statistically significant ($p = 0.048$). At month 3, there was a highly significant difference ($p = 0.007$) between the groups (fig. 3), but by month 6 there was no statistically significant difference in calcium $\times$ phosphate products between the groups. Similarly, following the end of the trial, there was no statistically significant difference between the two groups after 3 months (fig. 3) and also after 12 months: 3.78 ± 1.12 vs. 4.26 ± 1.27 mmol²/l², respectively.

Secondary Outcomes

The increased dietetic intervention did not significantly affect any other nutritional parameters during the trial, such as body mass index, serum potassium, corrected serum calcium, predialysis midweek serum urea, albumin, PTH, handgrip strength, and mid-upper-arm circumference (table 3).

There was no significant change found in the mean number of phosphate binder tablets prescribed to either group, or in the self-reported number of tablets taken by subjects during the trial. However, phosphate binders could be withdrawn in 2 patients, 1 in each group, as they were no longer required to achieve the serum phosphate target. A reduction in usage of aluminium-based phosphate binders was also observed, with 20 (29.9%) subjects on this binder type alone or in combination at baseline, compared to 14 (23.3%) subjects at month 6.

When patients were asked to score how often they failed to take their phosphate binders (i.e., 'rarely' or 'never' vs. '1 every couple of days' up to 'multiple times a day'), then there were no significant differences in self-reported patient compliance either between the intervention and control groups or within groups during the trial.

Discussion

The UK Renal Association Standards Committee predialysis target guideline for serum phosphate [16] and that of the North American K/DOQI [17] are similar at <1.8 mmol/l and 1.13–1.78 mmol/l, respectively. Prior to entry, all the patients recruited for this study persistently had serum phosphate concentrations above the UK Renal Association standard. During the study, there was an ini-

Table 3. Other nutritional parameters at the end of the study after 6 months (no significant differences between the groups)

	Intervention group (n = 30)	Control group (n = 30)
Body mass index (range)	26.9 (20.6–36.8)	23.5 (19.3–43.4)
Grip strength, kg	22.8 ± 9.2	19.4 ± 10.3
Mid-upper-arm circumference, cm	28.6 ± 3.0	28.0 ± 4.1
Potassium, mmol/l	5.2 ± 0.8	5.3 ± 0.6
Corrected calcium, mmol/l	2.4 ± 0.2	2.3 ± 0.2
Pre-dialysis urea, mmol/l	19.7 ± 5.7	21.0 ± 5.0
Albumin, mmol/l	40.8 ± 3.2	40.2 ± 3.9
Parathyroid hormone, pmol/l (range)	35.3 (2.9–110.4)	36.3 (1.3–202.6)

tial statistically significant improvement in serum phosphate control in the intervention group by month 3, and 52% of the intervention group achieved the Renal Association and/or K/DOQI phosphate targets, whereas only 23% of the control group attained the standards ($p = 0.003$). However, the reduction in serum phosphate was not sustained for the duration of the study, and similarly there was no difference on follow-up, once the study had been completed.

However, after adjusting for confounding variables, the treatment effect between the two groups on serum phosphate at month 3 ($p = 0.056$) was approaching significance, and may have been significant with greater patient numbers. Although statistical advice was provided in the initial study design, based on previously published work [12], it would appear that it was underpowered to detect a statistically significant difference in serum phosphate concentrations, once controlling for confounding variables, and these would need to be considered and incorporated into any future studies. One of the major differences in our study compared to previous reports was the improvement in serum phosphate in the control group, which acted as a confounding variable.

In terms of the calcium $\times$ phosphate product, reductions were observed in both groups during the trial, with the maximum difference between groups noted at month 3 ($p = 0.007$). The K/DOQI guidelines [17] suggest the product should be maintained $<4.4 \text{ mmol}^2/\text{l}^2$, as greater values have been linked to soft-tissue calcification and increased risk of both morbidity and death [1, 18]. After month 3, 58% of the intervention group achieved this standard, whereas only 37% of the control group attained the standard ($p = 0.01$). However, once again, these differences could not be sustained during the remainder of the intervention study, or on follow-up after study completion.

Our prospective randomized study found that intensive dietetic input was initially successful in the short term; however, it is unknown as to how much of the improvement was due to improved compliance with phosphate binders as opposed to dietary restrictions. While no significant differences in patient compliance were recorded in this trial, it is well recognized that compliance with phosphate binders is difficult to measure and often, as in this trial, relies on self-report. Future studies will need to address whether the improvement in serum phosphate control was due to a change in dietary phosphate intake or greater compliance taking prescribed phosphate binders or a combination of both and could include food diaries or a validated tool for measuring dietary phosphate intake and a pill count.

There was an unexpected improvement in the serum phosphate concentrations in the control group, despite all these patients initially having sustained serum phosphate concentrations above recommended guidelines. It is recognized that simply being part of a clinical trial, even as a placebo participant, may lead to an improvement. In addition, previous dietetic interventional trials have noted that there may be difficulty in truly separating the control group and blinding the intervention. In our study the patients were reviewed in their dialysis centre, and the research dietician observed that patients in the intervention group, whilst sitting and waiting for transport after their haemodialysis session, would discuss phosphate management strategies. There was thus potential for contamination, as patients in the control group may have been unduly influenced by contact with the intervention group. This highlights one of the difficulties of performing research in this population where chronic dialysis patients become well acquainted with each other.

The increased frequency of dietetic intervention did initially result in a lowering of serum phosphate concen-

trations and this could potentially translate into improved longer-term outcomes, such as mortality risk. Block et al. [1] observed that the relative risk of death for those haemodialysis patients with a serum phosphate level >2.1 mmol/l was 1.27 relative to those with lower concentrations of 0.78–2.1 mmol/l. As dietetic intervention is a relatively cheap therapy within the health care system, the improvements in serum phosphate control observed in this trial could potentially have more widespread benefits, both financially and in terms of health care outcomes, particularly if they could be sustained.

In addition, due to the improvement in serum phosphate control, the prescription of aluminium-based binders decreased during the trial, so reducing potential aluminium exposure and toxicity. In 2 subjects, phosphate control improved so much that phosphate binders could be withdrawn, and future research needs to focus on the costs and benefits of providing dietetic intervention as complementary to drug therapy on a more long-term, widespread scale.

In our study, the reduction in serum phosphate of 0.25 mmol/l observed at month 3 is somewhat less than that reported by Ashurst Ide and Dobbie [12] of 0.36 mmol/l. This may have well been due to differences in the populations studied, as in our study all patients recruited had persistently sustained hyperphosphataemia (mean >1.8 mmol/l for 3 months prior to entry into the study) and many were known to be non-compliant, whereas in the study by Ashurst Ide and Dobbie [12] the subjects only had to have one result >1.7 mmol/l to be included. Whereas our study included stable patients dialyzing for at least 6 months, Ashurst Ide and Dobbie [12] stated they included clinically stable patients, and one would assume that if new patients were included then their serum phosphate concentration would decline with increased dialysis adequacy. In addition, the interventions provided, patient education literature used, and other educational and/or motivational factors, or simply the impact of the specific research dietician, may have differed between the studies.

The initial improvements in serum phosphate control were not sustained in our intervention group and by month 6 the phosphate concentrations had stabilized. This would suggest that the intervention group may not have responded as favourably to the monthly dietetic sessions as they had initially. They may have also found it difficult to continue to comply for the 6-month study, as the diet itself is complex, with most dietary phosphate also being protein sources, so a fine balance is required in order to meet both of these nutritional goals. These

end-stage renal failure patients have complex therapeutic regimens, attending for dialysis treatment, prescription of numerous medications, coupled with dietary restrictions, and are often depressed. It is therefore understandable that long-term adherence may be difficult and depends on individual patient determination, ability, and motivation to co-operate.

Whenever there is a change in lifestyle or other behavioural intervention, it is well recognized that in the model of change [19], relapse is a normal part of the change cycle for some time, before people are then able to sustain changed behaviour. Depression, which is often not diagnosed, has also been identified as a complicating co-morbid diagnosis in this population group [20, 21], making counselling and behavioural change more difficult.

The results of our prospective randomized trial suggest that while traditional intensive dietetic counselling does appear to be initially successful in the short term, more innovative strategies may be needed to control hyperphosphataemia in the longer term. Possible solutions include group therapy, psychological self-management strategies, or preventative rather than reactive counselling at an earlier stage in disease progression. It is also plausible that different approaches may need to be individualized to different patients depending on their personality types, reasons for hyperphosphataemia, and specific compliance issues. Group-based strategies aimed at self-management, compliance, and improved motivation have proven successful in the diabetic population [22–24] and in some initial trials with dialysis patients [25], and warrants future investigation.

Acknowledgments

This work was supported by a grant from the British Renal Society. The authors also wish to acknowledge Dr. Richard Morris for his advice and assistance with the statistical methods. They would also like to thank the renal dieticians at Royal Free Hospital (Mrs. Jan Flint, Mrs. Ruth Devenish, Ms. Julia Hill, and Mrs. Sarah Crooks) for their guidance, support, and proofreading of drafts. The support provided by the entire renal department, particularly the nurses in the dialysis units, is also greatly appreciated. Finally, the patients who agreed to take part are also thanked for their cooperation.

References

- Block GA, Hulbert-Shearon TE, Levin NW, et al: Association of serum phosphorus and calcium $\times$ phosphate product with mortality risk in chronic hemodialysis patients: a national study. *Am J Kidney Dis* 1998;31:607–617.
- Goodman WG, Goldin J, Kuizon BD, et al: Coronary artery calcification in young adults with end-stage-renal disease who are undergoing dialysis. *N Engl J Med* 2000;342:1478–1483.
- Ganesh SK, Stack AG, Levin NW, et al: Association of elevated serum PO_4 , Ca $\times$ PO_4 product, and parathyroid hormone with cardiac mortality risk in chronic hemodialysis patients. *J Am Soc Nephrol* 2001;12:2131–2138.
- Raggi P, Boulay A, Chasan-Taber S, et al: Cardiac calcification in adult hemodialysis patients. A link between end-stage renal disease and cardiovascular disease? *J Am Coll Cardiol* 2002;39:695–701.
- Slatopolsky E, Dusso A, Brown AJ: The role of phosphorus in the development of secondary hyperparathyroidism and parathyroid cell proliferation in chronic renal failure. *Am J Med Sci* 1999;317:370–376.
- Yi H, Fukagawa M, Kurokawa K: Mild dietary phosphorus restriction directly prevents enhanced parathyroid hormone secretion and synthesis and proliferation of parathyroid cells in chronic renal failure (abstract). *J Am Soc Nephrol* 1992;3:703.
- Pei Y, Hercz G, Greenwood C, et al: Risk factors for renal osteodystrophy: a multivariate analysis. *J Bone Miner Res* 1995;10:149–156.
- Llach F, Yudd M: The importance of hyperphosphataemia in the severity of hyperparathyroidism and its treatment in patients with chronic renal failure. *Nephrol Dial Transplant* 1998;13(suppl 3):57–61.
- Block GA, Port FK: Re-evaluation of risks associated with hyperphosphatemia and hyperparathyroidism in dialysis patients: recommendations for a change in management. *Am J Kidney Dis* 2000;35:1226–1237.
- Haynes RB, Yao X, Degani A, et al: Interventions to enhance medication adherence. *Cochrane Database Syst Rev* 2005;2:CD000011.
- Durose CL, Holdsworth M, Watson V, et al: Knowledge of dietary restrictions and the medical consequences of noncompliance by patients on hemodialysis are not predictive of dietary compliance. *J Am Diet Assoc* 2004;104:35–41.
- Ashurst Ide B, Dobbie H: A randomized controlled trial of an educational intervention to improve phosphate levels in hemodialysis patients. *J Ren Nutr* 2003;13:267–274.
- Ford JC, Pope JF, Hunt AE, et al: The effect of diet education on the laboratory values and knowledge of hemodialysis patients with hyperphosphatemia. *J Ren Nutr* 2004;14:36–44.
- Gillis BP, Caggiula AW, Chiavacci AT, et al: Nutrition intervention program of the Modification of Diet in Renal Disease Study: a self-management approach. *J Am Diet Assoc* 1995;95:1288–1294.
- Locatelli F, Cannata-Andía JB, Drüeke TB, et al: Management of disturbances of calcium and phosphate metabolism in chronic renal insufficiency, with emphasis on the control of hyperphosphataemia. *Nephrol Dial Transplant* 2002;17:723–731.
- Renal Association: Treatment of Adults and Children with Renal Failure: Standards and Audit Measures, ed 3. London, Royal College of Physicians of London and the Renal Association, 2002.
- National Kidney Foundation: K/DOQI clinical practice guidelines for bone metabolism and disease in chronic kidney disease. *Am J Kidney Dis* 2003;42(suppl 3):S1–S201.
- Rufino M, García S, Jiménez A, et al: Heart valve calcification and calcium $\times$ phosphorus product in hemodialysis patients: analysis of optimum values for its prevention. *Kidney Int Suppl* 2003;85:115–118.
- Prochaska JO, DiClemente CC: Transtheoretical therapy: toward a more integrative model of change. *Psychother Theory Res Pract* 1982;19:276–288.
- Cukor D, Peterson RA, Cohen SD, et al: Depression in end-stage renal disease hemodialysis patients. *Nat Clin Pract Nephrol* 2006;2:678–687.
- Kimmel PL, Peterson RA: Depression in end-stage renal disease patients treated with hemodialysis: tools, correlates, outcomes, and needs. *Semin Dial* 2005;18:91–97.
- Deakin T, McShane CE, Cade JE, et al: Group based training for self-management strategies in people with type 2 diabetes mellitus. *Cochrane Database Syst Rev* 2005;2:CD003417.
- Steed L, Lankester J, Barnard M, et al: Evaluation of the UCL diabetes self-management programme (UCL-DSMP): a randomized controlled trial. *J Health Psychol* 2005;10:261–276.
- Norris SL, Engelgau MM, Narayan KM: Effectiveness of self-management training in type 2 diabetes: a systematic review of randomized controlled trials. *Diabetes Care* 2001;24:561–587.
- Shaw-Stuart NJ, Stuart A: The effect of an educational patient compliance program on serum phosphate levels in patients receiving hemodialysis. *J Ren Nutr* 2000;10:80–84.